

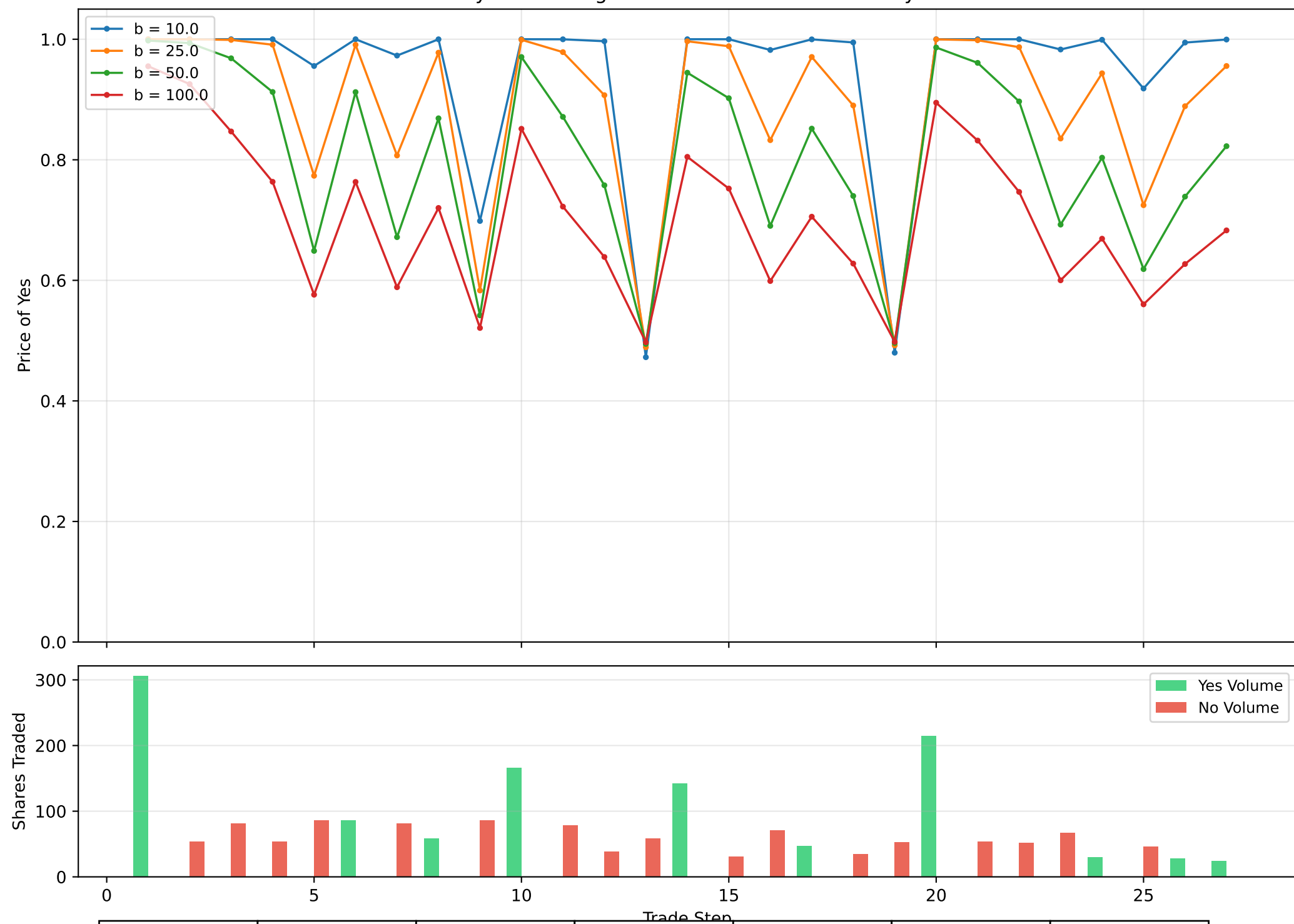
LMSR Rug Pull & Long Trend Analysis

Impact of Liquidity Parameter (b) on Price Paths

b values analyzed: [10.0, 25.0, 50.0, 100.0]

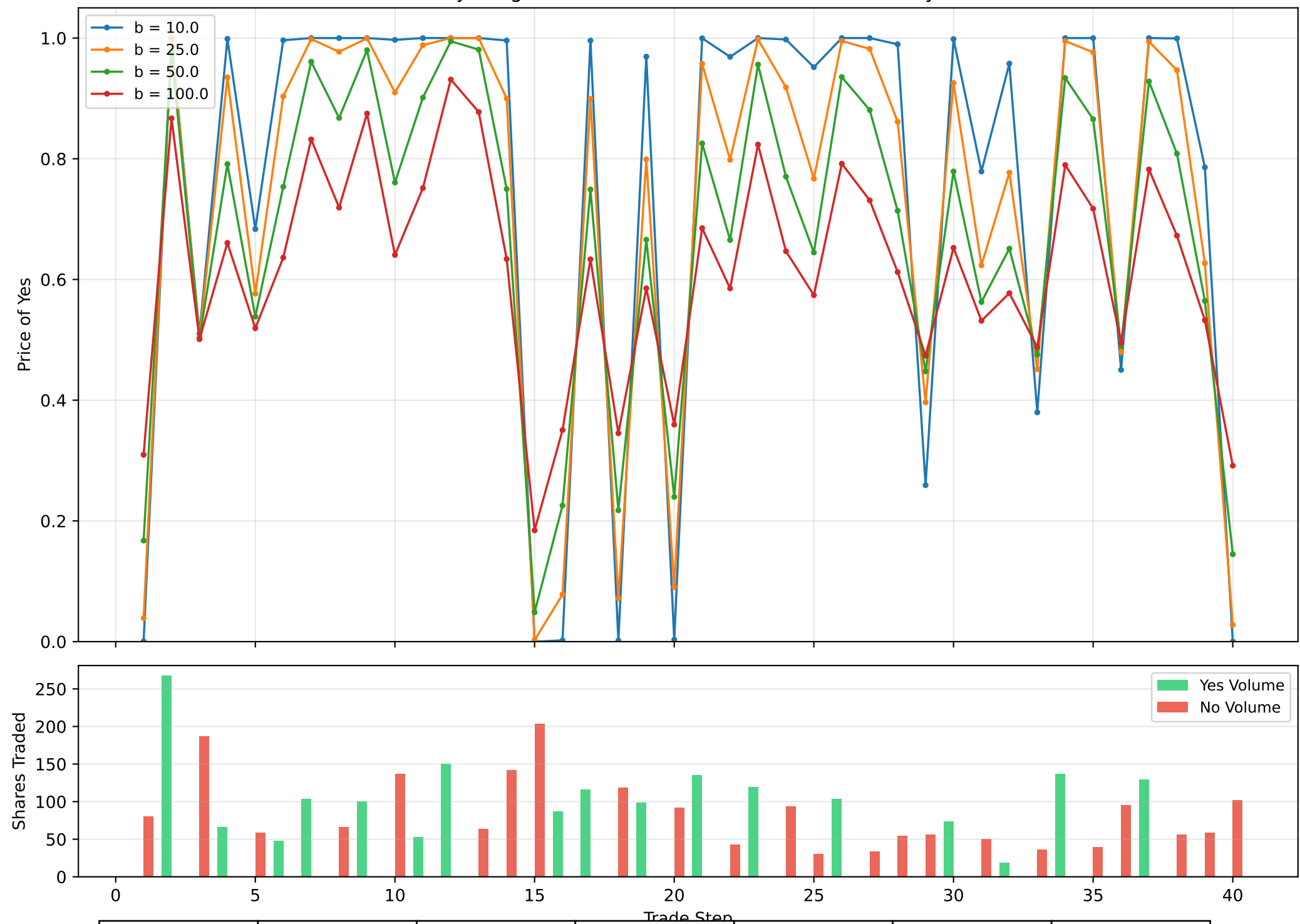
Generated from example trade histories in examples/trade_histories/

Kelly Classic Rug Pull — Price Path + Volume by b



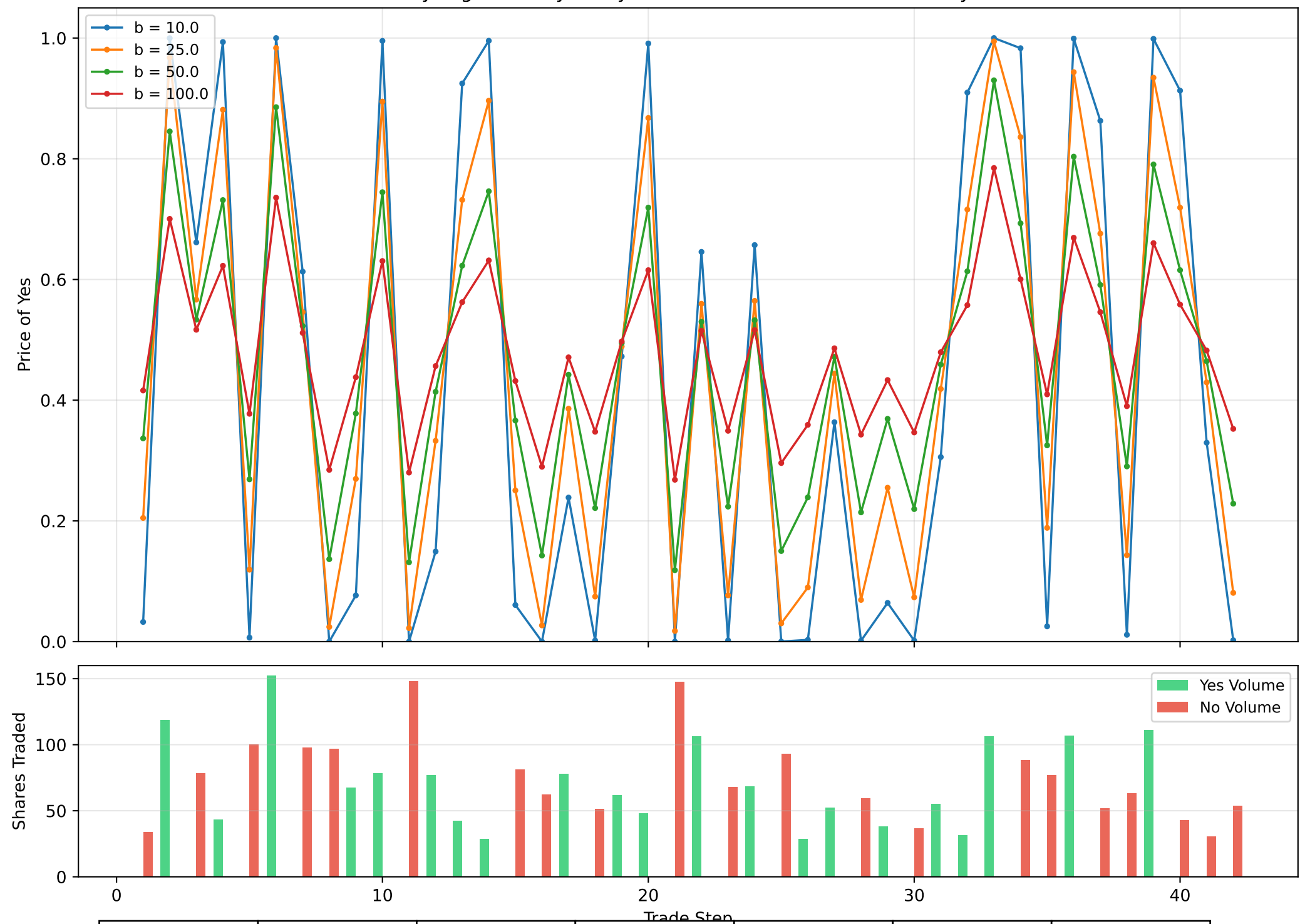
b	QV	TV	Roughness	Step Vol	Max Step	Final p-0.5
10.0	1.2891	3.0711	2.705	0.2227	0.5275	0.4995
25.0	1.5043	4.6368	3.781	0.2405	0.5078	0.4556
50.0	1.3239	4.7864	4.160	0.2257	0.4903	0.3226
100.0	0.6939	3.5757	4.293	0.1634	0.3967	0.1829

Kelly Long Gradual Trend — Price Path + Volume by b



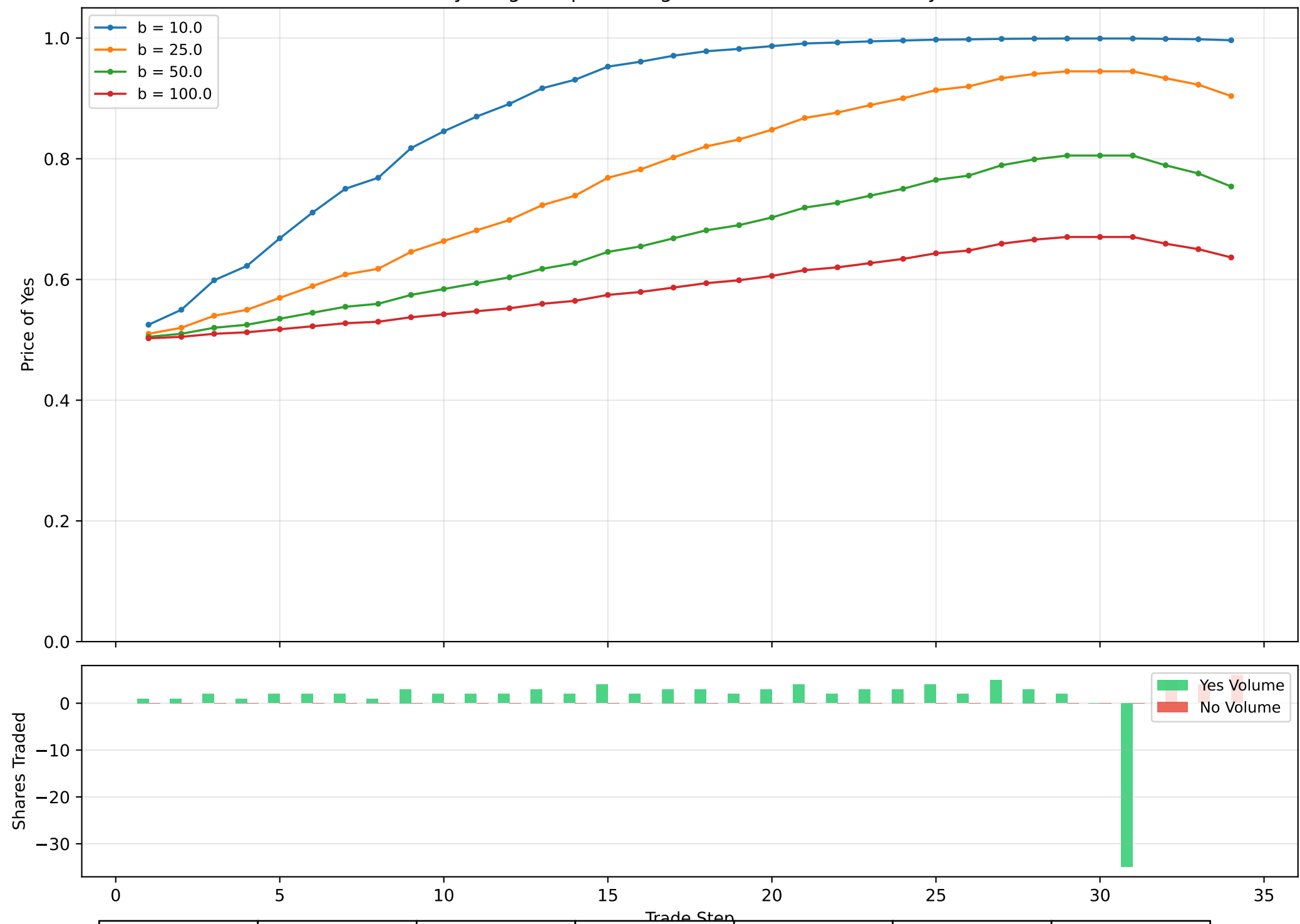
b	QV	TV	Roughness	Step Vol	Max Step	Final p-0.5
10.0	10.6574	13.8700	4.249	0.5227	0.9997	0.4999
25.0	7.7232	13.1798	4.743	0.4450	0.9605	0.4721
50.0	4.4142	11.0218	5.246	0.3364	0.8096	0.3552
100.0	1.9107	7.5397	5.455	0.2213	0.5574	0.2085

Kelly High-Activity Noisy Market — Price Path + Volume by b



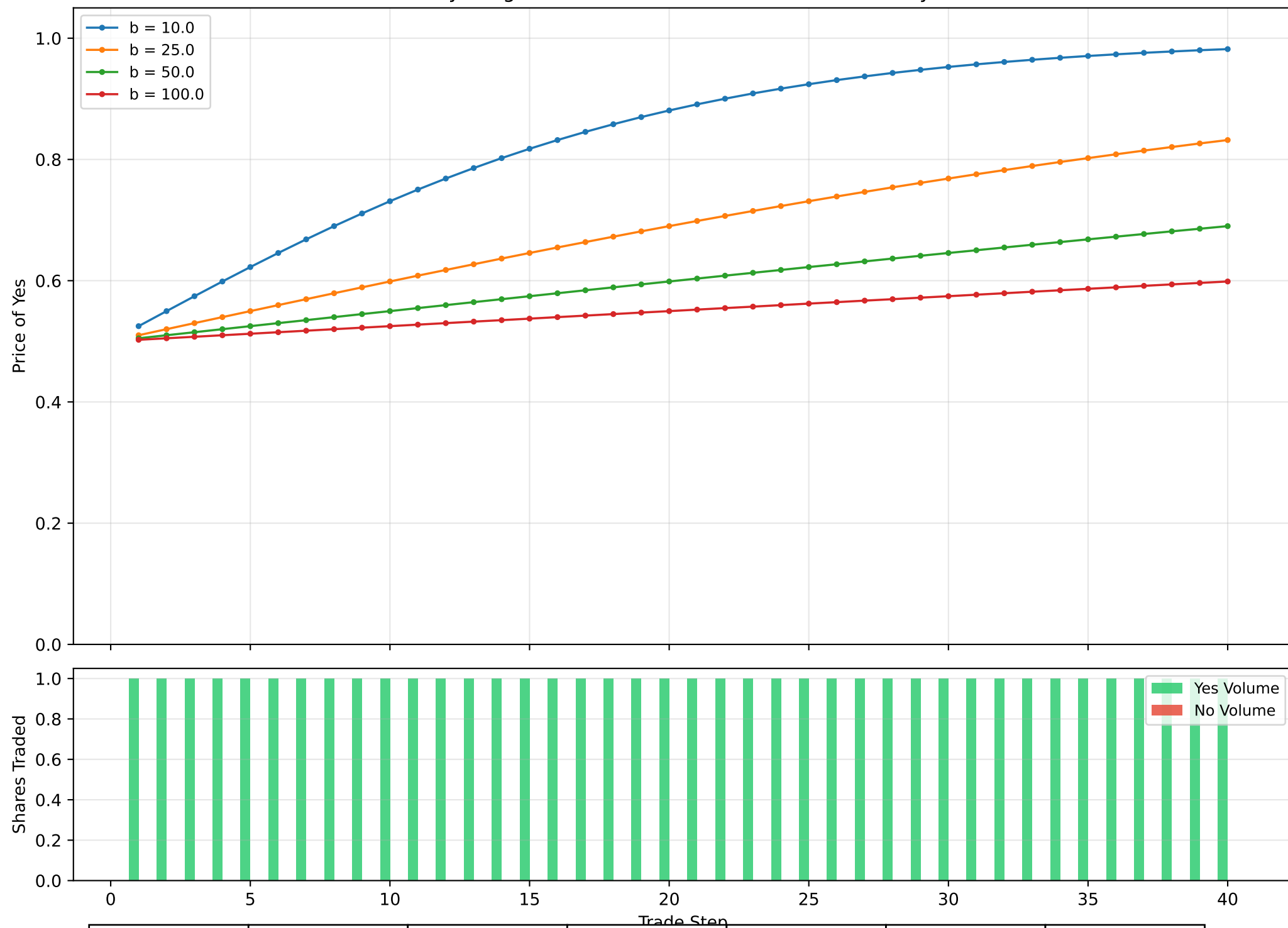
b	QV	TV	Roughness	Step Vol	Max Step	Final p-0.5
10.0	15.0430	20.4239	5.266	0.6057	0.9952	0.4977
25.0	9.7778	17.9935	5.754	0.4883	0.8723	0.4192
50.0	4.5724	12.5671	5.877	0.3340	0.6166	0.2714
100.0	1.4909	7.1770	5.878	0.1907	0.3581	0.1475

Very Long Pump and Rug — Price Path + Volume by b



b	QV	TV	Roughness	Step Vol	Max Step	Final p-0.5
10.0	0.0152	0.4771	3.867	0.0215	0.0491	0.4963
25.0	0.0084	0.4758	5.193	0.0160	0.0296	0.4038
50.0	0.0045	0.3516	5.252	0.0117	0.0216	0.2540
100.0	0.0015	0.2018	5.154	0.0068	0.0137	0.1365

Very Long Gradual Trend — Price Path + Volume by b



b	QV	TV	Roughness	Step Vol	Max Step	Final p-0.5
10.0	0.0077	0.4570	5.211	0.0140	0.0248	0.4820
25.0	0.0027	0.3220	6.160	0.0084	0.0100	0.3320
50.0	0.0009	0.1850	6.238	0.0047	0.0050	0.1900
100.0	0.0002	0.0962	6.244	0.0025	0.0025	0.0987

Key Observations

- Lower b values cause much larger and faster price swings.
- In rug-pull scenarios, low b makes the price run much higher before the dump.
- High b markets are far more resilient to large single trades (less slippage).
- In long gradual trends, low b can push prices close to 1.0, while high b stays moderate.
- Rug pulls with fake volume (coordinated accounts) are more effective at low b .

Recommendation for internal tools:

- Use moderate-to-high b (40-100) if you want stable prices and less manipulation risk.
- Use lower b (10-25) only when you want prices to be very responsive to new information.